



RAJALAKSHMI ENGINEERING COLLEGE

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FREELANCER HUB : LEARN AND GROW

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BONAFIDE CERTIFICATE

Certified that this Project titled “**FREELANCER HUB : LEARN AND GROW**” is the Bonafide work of “**ELUMALAI B. (2116230701084)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

Submitted for the End semester practical examination to be held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

In today's rapidly evolving digital economy, freelancing platforms have become essential for connecting skilled professionals with companies seeking flexible and efficient workforce solutions. However, most existing systems primarily focus on project hiring and lack integrated collaboration, productivity tracking, and freelancer management capabilities. To address these limitations, *Freelancer Hub* is developed as an Android-based mobile application using Kotlin in Android Studio, **aimed at creating a unified platform for freelancing, recruitment, communication, and productivity enhancement**. The application introduces a unique dual-role system where freelancers are not only able to work on projects but also recruit and manage sub-freelancers for delegated tasks. **Companies are provided with a dedicated interface to post projects, review proposals, shortlist candidates, and communicate directly** through an integrated real-time chat system. The platform also includes project lifecycle management features that help users track project status, deadlines, milestones, and task completion efficiently. To improve user engagement and work-life balance, the system integrates additional productivity-oriented features such as earnings tracking, daily stress monitoring, activity analysis, and mini interactive games that help reduce mental fatigue and maintain active participation. The application supports customizable themes, light/dark modes, and personalized user interface elements for enhanced accessibility and user experience. The entire system is powered by a single embedded SQLite database file, ensuring lightweight operation, fast local data access, secure offline functionality, and efficient relational data management without dependency on external servers. Freelancer Hub demonstrates how modern mobile technologies can integrate freelancing, recruitment, communication, financial tracking, and productivity management into a single scalable and user-friendly mobile application.

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CHAPTER 1 INTRODUCTION

1.1 GENERAL

Freelancer Hub is an Android-based mobile application developed using Kotlin in Android Studio, designed to provide a unified platform for freelancing, recruitment, communication, and productivity management. The application connects freelancers and companies through a digital workspace where companies can post projects, review proposals, and recruit suitable candidates, while freelancers can apply for jobs, manage projects, and collaborate with other freelancers. A unique feature of the system is its dual-role functionality, which allows freelancers to both work on projects and recruit sub-freelancers for delegated tasks.

The application integrates advanced features such as real-time chat, project tracking, earnings monitoring, daily stress analysis, and mini interactive games to improve user engagement and productivity. Additional customization options including light/dark mode and personalized UI elements enhance accessibility and user experience. The system uses a single embedded SQLite database file for secure, lightweight, and efficient offline data management, making Freelancer Hub a scalable and user-friendly mobile application.

1.2 SCOPE OF THE WORK

The scope of this project focuses on the design and development of *Freelancer Hub*, a comprehensive mobile application that integrates freelancing, recruitment, communication, and productivity management into a single platform. The system enables companies to post projects, review proposals, recruit freelancers, and communicate efficiently, while freelancers can apply for jobs, manage projects, delegate tasks, and collaborate with

sub-freelancers through a unique dual-role functionality. The application also includes additional productivity-oriented features such as project lifecycle tracking, earnings monitoring, daily stress analysis, mini interactive games, customizable themes, and real-time chat support. The system is developed using Kotlin in Android Studio and utilizes a single embedded SQLite database file for secure, lightweight, and offline-capable data management.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The primary aim of this project is to design and develop *Freelancer Hub*, a comprehensive Android-based mobile application that integrates freelancing, recruitment, communication, and productivity management into a single unified platform. The application is intended to simplify project hiring, improve collaboration between freelancers and companies, and provide a flexible environment for managing projects and delegated tasks efficiently.

The main objectives of the project are:

- To develop a dual-role system where freelancers can both work on projects and recruit sub-freelancers.
- To provide companies with an efficient platform for posting jobs, reviewing proposals, and selecting suitable candidates.
- To implement a real-time chat system for seamless communication between users.
- To integrate productivity-oriented features such as project tracking, earnings monitoring, stress analysis, and activity management.
- To enhance user experience through customizable themes, light/dark modes, and interactive UI elements.
- To utilize a single embedded SQLite database file for secure, lightweight, and offline-capable data management.
- To create a scalable and user-friendly freelancing ecosystem using modern mobile development technologies.

CHAPTER 2

LITERATURE SURVEY

Various research endeavors concerning freelancing platforms, mobile productivity applications, and collaborative workforce management systems emphasize an increasing demand for integrated digital environments that facilitate efficient communication, project management, and flexible recruitment methodologies. Traditional freelancing applications predominantly focus on job posting, proposal submission, and candidate hiring, but frequently offer restricted support for task delegation, productivity monitoring, team collaboration, and user engagement. Recent academic studies underscore the significance of real-time communication systems, project lifecycle tracking, and streamlined proposal management in enhancing workflow transparency, recruitment efficiency, and user satisfaction. Research on Android-based mobile applications developed using **Kotlin** in **Android Studio** demonstrates superior scalability, modularity, and performance within modern mobile ecosystems.

Furthermore, several investigations into offline-capable mobile applications indicate that embedded databases, specifically **SQLite**, provide a lightweight, secure, and rapid local data storage solution without necessitating external servers or continuous internet connectivity. Productivity-oriented applications that integrate features such as earnings tracking, task management, activity monitoring, stress analysis, and gamification techniques have demonstrated marked improvements in productivity, user retention, and mental well-being. Research also indicates that customizable user interfaces, light/dark themes, and personalized dashboard elements significantly enhance accessibility and the overall user experience. These collective studies establish the technical and conceptual foundation for developing *Freelancer Hub*, an intelligent and scalable mobile application designed to integrate freelancing, recruitment, communication, productivity management, and user engagement .

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

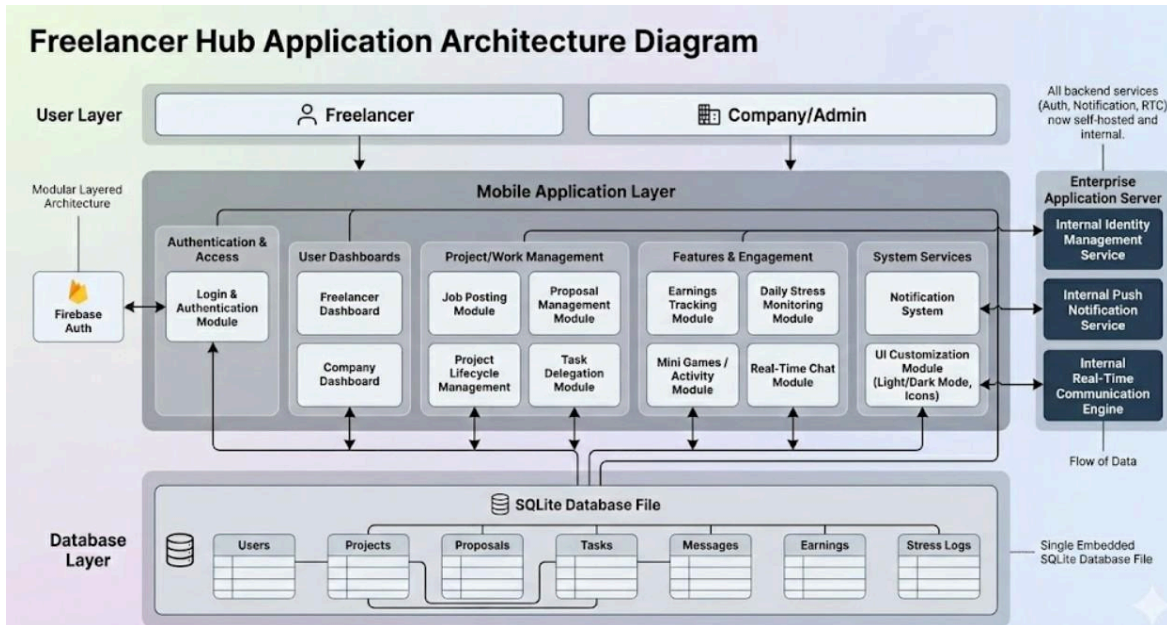


Fig 3.1.1 – Architecture Diagram

The structural framework of *Freelancer Hub* is predicated on a modular, multi-layered design philosophy, meticulously optimized for robust scalability, streamlined communication, and superior mobile performance. The system architecture is bifurcated into a User Layer, Application Layer, Service Layer, and Database Layer, which collectively facilitate integrated freelancing and recruitment operations. Within the application layer, specialized modules govern authentication, freelancer and company dashboards, project lifecycle management, proposal processing, real-time communication, financial tracking, stress analysis, and interface personalization. The core data infrastructure relies on a single embedded **SQLite** database. This local repository efficiently manages relational tables for users, projects, proposals, tasks, messages, earnings, and comprehensive activity logs, ensuring secure, offline-capable data management and an

organized workflow within a scalable mobile ecosystem.

3.1 USE CASE DIAGRAM

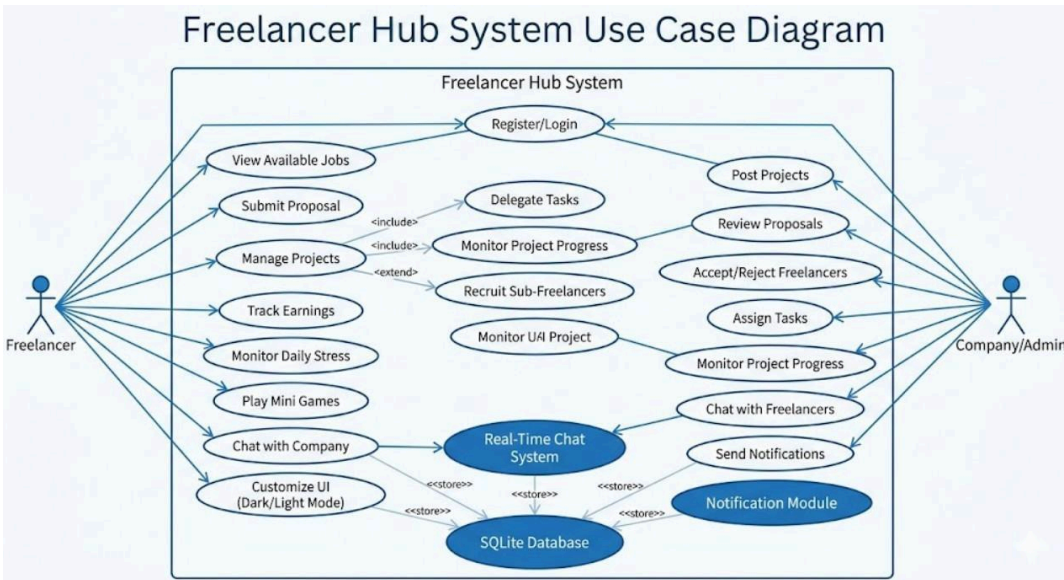


Fig 3.2.1 – Use Case Diagram

The Freelancer Use Case Diagram illustrates the interactions between freelancers and the Freelancer Hub system. Freelancers can register/login, browse available projects, submit proposals, manage assigned tasks, recruit sub-freelancers, track earnings, monitor stress levels, communicate with companies through real-time chat, and customize application settings such as light/dark mode. The diagram highlights how freelancers interact with different modules of the system to efficiently manage projects, collaboration, productivity, and communication within the platform.

3.2 HARDWARE SPECIFICATIONS

The Freelancer Hub mobile application is designed to operate efficiently on standard Android smartphones and computing devices without requiring specialized hardware components. The system architecture supports modules such as project management, real-time communication, earnings tracking, stress monitoring, and productivity features while maintaining smooth performance on mid-range mobile devices. The following hardware specifications are recommended for optimal performance:

- **Processor:** Quad-core processor (Qualcomm Snapdragon 660 or higher / MediaTek Helio series or equivalent)
- **RAM:** Minimum 4 GB RAM recommended for smooth multitasking
- **Storage:** Minimum 64 GB internal storage for application data and media handling
- **Display:** Touchscreen display with HD resolution support
- **Graphics:** Integrated mobile GPU support
- **Operating System:** Android 8.0 (Oreo) or higher
- **Network Requirements:** Wi-Fi or mobile internet connection for real-time chat and notifications
- **Additional Requirements:** Microphone support for communication features and notification support for alerts and reminders

These specifications ensure efficient execution of application modules, smooth user interaction, secure local database handling using SQLite, and seamless communication between freelancers and companies within the platform.

3.4 SOFTWARE SPECIFICATIONS

The Freelancer Hub mobile application is developed using a modern Android-based software architecture to ensure scalability, smooth performance, and efficient user interaction across different mobile devices. The application integrates modules such as freelancing management, recruitment handling, real-time communication, productivity tracking, earnings monitoring, and user engagement features within a unified platform. The software stack used for the development of the system includes the following technologies:

- **Frontend Framework:**
Kotlin with Android Studio for developing the Android mobile application and user interface components
- **UI/UX Design Tools:**
XML Layouts and Material Design for responsive, interactive, and user-friendly interfaces
- **Database:**
SQLite embedded database for lightweight, secure, and offline-capable local data storage
- **Real-Time Communication:**
Firebase services and Android networking components for chat and notification management
- **Data Handling & Processing:**
Kotlin Standard Library and Android SDK components for efficient data processing and application management
- **Development Environment:**
Android Studio IDE with Gradle build system for application development and testing

These software technologies ensure efficient project management, secure communication, smooth UI performance, and scalable mobile application development within the Freelancer Hub platform.

4. PROPOSED SYSTEM

The proposed system, *Freelancer Hub*, is an Android-based mobile application designed to integrate freelancing, recruitment, communication, and productivity management into a single platform. The system allows companies to post projects, review proposals, assign tasks, and communicate with freelancers, while freelancers can apply for projects, manage work, collaborate with others, and recruit sub-freelancers through a dual-role functionality. This improves flexibility, collaboration, and overall workflow efficiency.

The application also includes advanced features such as real-time chat, project tracking, earnings monitoring, stress analysis, mini interactive games, and customizable UI themes including light/dark mode. The entire system uses a single embedded SQLite database file for secure, lightweight, and offline-capable data management, while Firebase services are used for authentication and notifications. The proposed system aims to provide a scalable, user-friendly, and productivity-oriented freelancing ecosystem.

FEATURES OF THE PROPOSED SYSTEM

1. Dual-role freelancer and recruitment system
2. Project posting and proposal management
3. Real-time chat and notifications
4. Project lifecycle and task tracking
5. Earnings monitoring system
6. Daily stress analysis and activity tracking
7. Mini interactive games for engagement
8. Light/Dark mode and UI customization
9. SQLite-based offline data storage
10. Secure authentication and scalable architecture

5. MODULE DESCRIPTION

The *Freelancer Hub – A Dual Role Freelancing & Recruitment Mobile Application* is divided into five major modules that collectively manage freelancing, recruitment, communication, productivity tracking, and data handling within the platform. Each module is designed to provide efficient functionality while ensuring a smooth and user-friendly mobile experience.

User & Authentication Module

This module handles user registration, login, and profile management for both freelancers and companies. Users can securely create accounts, authenticate themselves, and access role-based functionalities within the application. The system uses Firebase Authentication to ensure secure login management and controlled access to different features. User details such as profile information, skills, project history, and account settings are maintained within the application for personalized interaction and efficient workflow management.

Freelancing & Recruitment Module

The Freelancing and Recruitment Module forms the core functionality of the application. Companies can post projects, review freelancer proposals, shortlist candidates, and assign tasks efficiently. Freelancers can browse available projects, submit proposals, manage ongoing work, and collaborate with sub-freelancers through the platform's dual-role system. The module also supports project lifecycle management including deadlines, milestones, task delegation, and completion tracking, improving collaboration and workflow efficiency between users.

Communication & Notification Module

This module enables real-time interaction between freelancers and companies

through an integrated chat system. Users can exchange messages, discuss project requirements, receive updates, and communicate efficiently within the platform. Notification services provide instant alerts regarding project updates, proposal acceptance, new messages, deadlines, and task changes. This module helps reduce communication delays and improves coordination among users.

Productivity & Activity Management Module

The Productivity Module is designed to improve user engagement and work-life balance through additional productivity-oriented features. It includes earnings tracking, daily stress monitoring, activity analysis, and mini interactive games that help reduce mental fatigue and maintain user activity. Freelancers can monitor project-wise earnings and productivity levels while also accessing customizable UI features such as light/dark mode and personalized dashboard elements for enhanced user experience.

Database Management Module

The Database Management Module handles all application data storage and retrieval processes using a single embedded SQLite database file. Core information including users, projects, proposals, tasks, messages, earnings, and activity logs are securely maintained within the SQLite architecture. This approach ensures lightweight operation, fast local data access, secure offline functionality, and efficient relational data management without dependency on external servers, making the application scalable and mobile-friendly.

5.1 KEY BENEFITS OF PROPOSED SYSTEM

Enhanced Freelancing & Recruitment Management:

The application provides an efficient platform for companies to post projects,

review proposals, recruit freelancers, and manage project workflows seamlessly. Freelancers can also recruit sub-freelancers, improving collaboration and flexibility.

Real-Time Communication:

The integrated chat and notification system enables smooth communication between freelancers and companies, reducing delays and improving project coordination and decision-making.

Improved Productivity & User Engagement:

Features such as earnings tracking, stress monitoring, activity analysis, and mini interactive games help users maintain productivity, reduce mental fatigue, and improve overall engagement within the platform.

Secure & Lightweight Data Management

The system uses a single embedded SQLite database file for secure, lightweight, and offline-capable data storage, ensuring fast data access and efficient application performance without dependency on external servers.

User-Friendly & Customizable Interface:

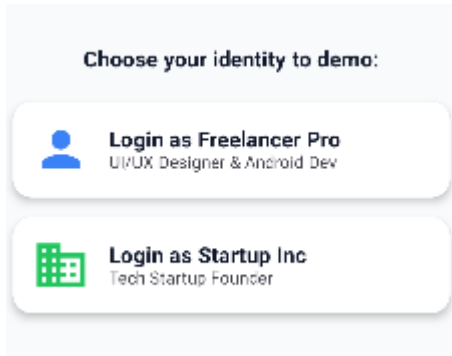
The application supports responsive UI layouts, light/dark mode, customizable icons, and personalized dashboard elements, providing a better user experience and accessibility across different devices.

Scalable & Flexible Architecture:

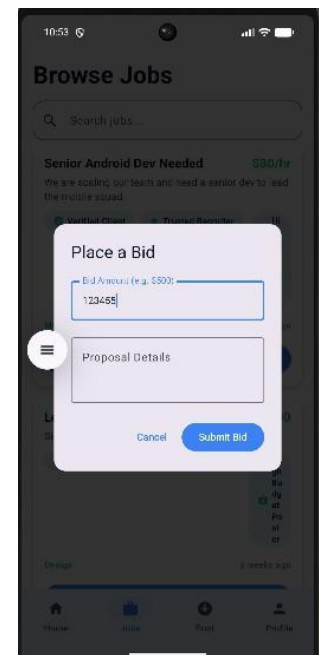
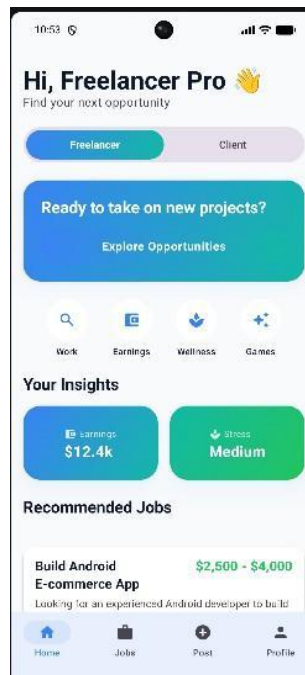
The modular system architecture allows easy future enhancements such as AI-based recommendations, payment gateway integration, analytics dashboards, and cross-platform support

CHAPTER-6

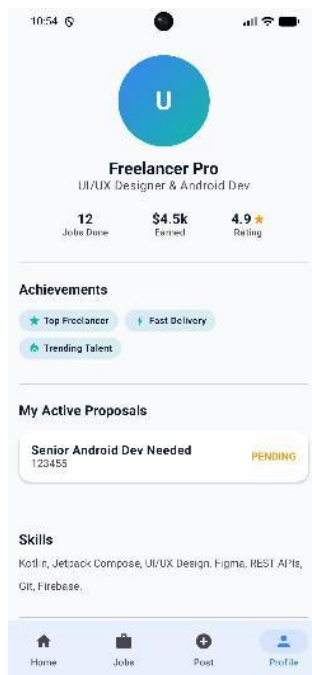
OUTPUT



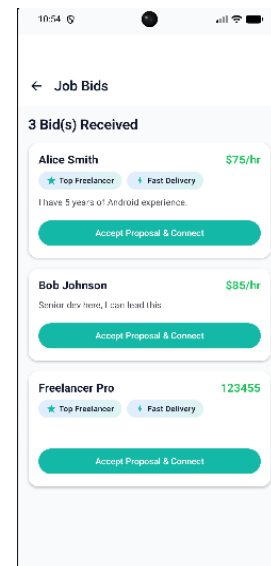
6.1 Login as roles



6.2 Freelancer portal and place bid



6.1 Freelancer profile



6.3 Accept and start



CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

The *Freelancer Hub* application provides a unified mobile platform for freelancing, recruitment, communication, and productivity management within a single system. By integrating features such as project posting, proposal management, real-time chat, task delegation, earnings tracking, stress monitoring, and activity-based engagement, the application improves collaboration and workflow efficiency between freelancers and companies. The use of a single embedded SQLite database file ensures lightweight, secure, and offline-capable data management, while customizable UI features enhance accessibility and user experience. Overall, the system demonstrates how modern mobile technologies can be effectively utilized to build a scalable and user-friendly freelancing ecosystem.

Future Enhancements:

1. AI-based freelancer and project recommendation system
2. Integrated payment gateway and transaction management
3. Cross-platform support for web and iOS applications
4. Advanced analytics dashboard for productivity and earnings insights
5. Video calling and meeting integration for communication
6. Cloud synchronization and automated backup support
7. Blockchain-based secure contract and payment verification
8. Multi-language support for global accessibility



CHAPTER 8

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